PRACTICAL NO 08

Name : Pranjal somatkar class: SY-B Roll no. : B-48

AIM: Implement and analyze various methods of studying correlation and regression including Karl Pearson's coefficient and simple linear regression.

Theory

1. What is Correlation? How is it different from Regression?

Correlation

Correlation is a statistical measure that describes the **strength and direction of a relationship** between two variables.

- It tells whether variables move **together** or **in opposite directions**
- Value lies between **-1 and +1**

Types of Correlation

Type	Value Range	Meaning
Positive Correlation	0 to +1	Both variables increase together
Negative Correlation	0 to -1	One increases, other decreases
Zero Correlation	0	No relationship

Example

- Study hours ↑ → Marks ↑ → Positive correlation
- Price ↑ → Demand ↓ → Negative correlation

Regression

Regression is a statistical method used to **predict the value of one variable based on another**.

- It shows a **functional relationship**
- Used for **forecasting and decision-making**

Correlation vs Regression (Detailed Table)

Feature	Correlation	Regression
Purpose	Measure relationship	Predict outcome
Output	Single value (r)	Equation (y = mx + c)
Direction	No cause-effect	Shows dependency
Variable roles	No distinction	Independent & dependent
Usage	Understanding relation	Prediction and modeling
Units	Unit-free	Depends on variables

Key Insight

- Correlation tells **"how strongly variables are related"**
- Regression tells **"how one variable affects another"**

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2. What is Simple Linear Regression? Explain with Example

Definition

Simple Linear Regression is a method used to model the relationship between **one independent variable (X)** and **one dependent variable (Y)** using a straight line.

Mathematical Equation

e

[y = mx + c]

Where:

- **y** → dependent variable
- **x** → independent variable
- **m** → slope (rate of change)
- **c** → intercept

Interpretation

- **Slope (m)** → how much y changes when x increases
- **Intercept (c)** → value of y when x = 0

Example

Study Hours (X)	Marks (Y)
1	40
2	50
3	60
4	70

Regression equation: [Marks = 10 \times Hours + 30]

Graphical Representation

- Points plotted on graph
- Best-fit straight line drawn
- Used for prediction

Applications

Field	Use
Education	Predict marks
Business	Sales forecasting
Finance	Risk prediction
Healthcare	Disease analysis

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3. How do you calculate Pearson Correlation in Python?

Karl Pearson's Correlation Coefficient

It measures the **linear relationship between two variables**.

Formula

[r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}]

Python Implementation

```
import pandas as pd

# Sample dataset
data = {
    'hours': [1, 2, 3, 4, 5],
    'marks': [40, 50, 60, 70, 80]
}

df = pd.DataFrame(data)

# Method 1: Using pandas
r = df['hours'].corr(df['marks'])
print("Pearson Correlation:", r)

# Method 2: Using numpy
```

```
import numpy as np
r_np = np.corrcoef(df['hours'], df['marks'])[0,1]
print("Correlation using numpy:", r_np)
```

Output Interpretation

Value of r	Meaning
+1	Perfect positive
-1	Perfect negative
0	No relationship

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4. What is Multicollinearity? Does it affect Simple Regression?

Definition

Multicollinearity occurs when **two or more independent variables are highly correlated** with each other.

Example

- Income and education level both predicting spending
- If they are strongly related → multicollinearity occurs

Effects of Multicollinearity

Effect	Explanation
Unstable coefficients	Small changes affect results
Reduced interpretability	Hard to identify important variable
Inflated variance	Less reliable estimates

Does it affect Simple Regression?

No, it does not affect simple regression

Because:

- Simple regression has **only one independent variable**
- Multicollinearity occurs only in **multiple regression**

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5. A dataset shows strong correlation but poor prediction accuracy. Why?

This is a common real-world issue.

Reasons Explained

Reason	Explanation
Non-linear relationship	Correlation measures only linear patterns
Outliers	Extreme values distort results
Overfitting/Underfitting	Model not properly trained
Small dataset	Not enough data
Missing variables	Important factors ignored
Noise in data	Random errors affect prediction

Example

- Data shows strong correlation
- But relationship is **curved (not linear)** → Regression line fails to predict accurately

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Key Insight

- High correlation ≠ Good prediction
- Model selection matters

Python Code (Correlation + Regression on Iris Dataset)

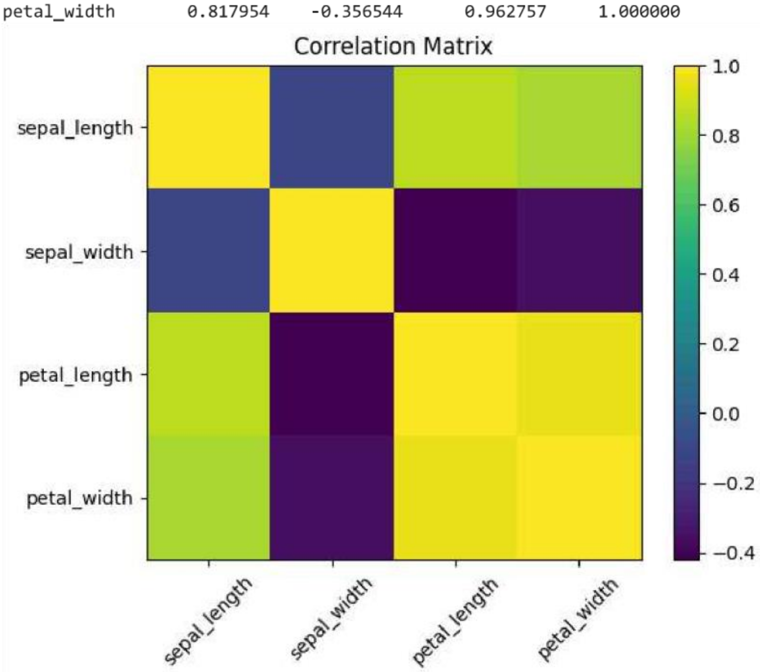
```
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Load dataset
df = pd.read_csv('/content/iris[1].csv')

# Display first rows
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

1. Correlation Matrix (All Features)

```
# Compute correlation matrix
corr_matrix = df.corr(numeric_only=True)

print("Correlation Matrix:\n", corr_matrix)

# Plot heatmap (simple)
plt.figure()
plt.imshow(corr_matrix, interpolation='nearest')
plt.colorbar()
plt.title("Correlation Matrix")
plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns, rotation=45)
plt.yticks(range(len(corr_matrix.columns)), corr_matrix.columns)
plt.show()
```

```
Correlation Matrix:
               sepal_length  sepal_width  petal_length  petal_width
sepal_length      1.000000    -0.109369      0.871754     0.817954
sepal_width      -0.109369     1.000000     -0.420516    -0.356544
petal_length      0.871754    -0.420516      1.000000     0.962757
petal_width       0.817954    -0.356544      0.962757     1.000000
```



2. Pearson Correlation (Two Variables)

```
# Example: Sepal Length vs Petal Length
x = df['sepal_length']
y = df['petal_length']
```

```
r = x.corr(y)
print("Pearson Correlation (sepal_length vs petal_length):", r)
```

```
Pearson Correlation (sepal_length vs petal_length): 0.8717541573048718
```

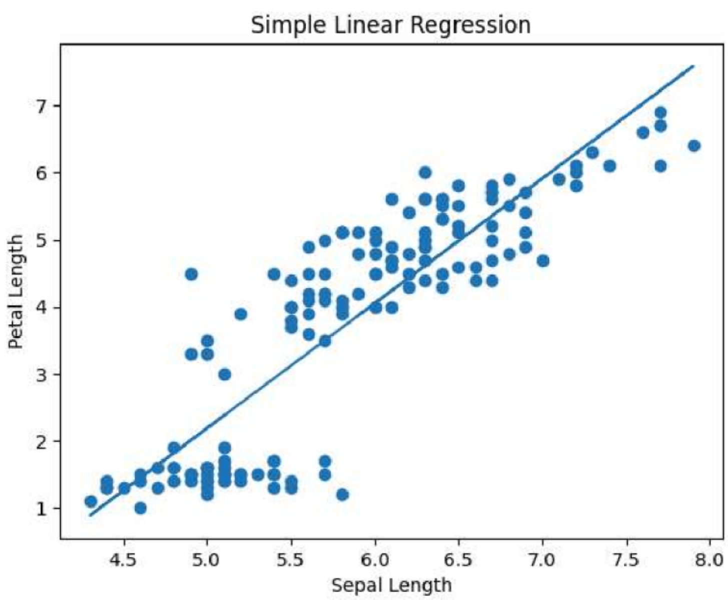
3. Simple Linear Regression

```
# Independent and dependent variables
X = df[['sepal_length']]   # independent
Y = df['petal_length']     # dependent

# Create model
model = LinearRegression()
model.fit(X, Y)

# Predictions
Y_pred = model.predict(X)

# Print coefficients
print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
```

```
Slope (m): 1.8575096654214456
Intercept (c): -7.0953814782793145
```

4. Plot Regression Line

```
plt.figure()
plt.scatter(X, Y)
plt.plot(X, Y_pred)
plt.title("Simple Linear Regression")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()
```



5. Check Prediction Accuracy (R^2 Score)

```
score = model.score(X, Y)
print("R² Score:", score)
```

```
R² Score: 0.7599553107783261
```

6. Demonstrating Poor Prediction

```
X2 = df[['sepal_width']]
Y2 = df['petal_length']

model2 = LinearRegression()
```

```
model2 = LinearRegression()
model2.fit(X2, Y2)

print("R² (weak relation):", model2.score(X2, Y2))
```

```
R² (weak relation): 0.17683378733246502
```

Conclusion

In this assignment, we studied the concepts of **correlation and regression** and their importance in data analysis. Correlation helps in measuring the **strength and direction of the relationship** between variables, while regression is used to **predict one variable based on another**.

We implemented **Karl Pearson's correlation coefficient** to understand linear relationships and applied **simple linear regression** to model and predict data. Using the dataset, we observed how different variables are related and how regression provides a best-fit line for prediction.

We also learned that **multicollinearity does not affect simple regression**, as it involves only one independent variable. Additionally, we understood that **strong correlation does not always guarantee accurate predictions**, as factors like non-linearity, outliers, and missing variables can impact results.

Overall, this assignment helped in understanding how correlation and regression are essential tools in statistics, machine learning, and real-world decision-making.